

❏ 1. Overview of Windows Operating System

❏ What is Windows OS?

Windows is a **graphical operating system** developed by **Microsoft**. It provides a **user interface (UI)** that allows users to interact with files, applications, hardware, and the internet. Windows supports **multitasking**, **virtual memory**, **security features**, and a **broad hardware base**, making it a dominant OS in personal computing and enterprise environments.

❏ Historical Evolution

Version	Year	Key Features
Windows 1.0	1985	First GUI for MS-DOS
Windows 95	1995	Start Menu, Plug and Play, multitasking
Windows XP	2001	Stability, improved UI, wide adoption
Windows 7	2009	Performance boost, better UAC, Aero
Windows 10	2015	Universal apps, Cortana, better security
Windows 11	2021	Modern UI, centered taskbar, Android app support

☒ Architecture of Windows Operating System

Windows OS uses a **hybrid kernel** architecture. Its components are:

1. Hardware Abstraction Layer (HAL)

- Isolates hardware differences from the OS
- Ensures portability

2. Kernel

- Manages low-level operations (thread scheduling, interrupt handling)
- Supports preemptive multitasking and SMP

3. Executive Services

- Includes core subsystems:
 - Object Manager
 - Memory Manager
 - Process Manager
 - I/O Manager
 - Security Reference Monitor

4. Device Drivers

- Interface between hardware and OS

5. Win32 Subsystem

- Provides API for GUI-based applications

6. User Mode

- Runs applications and subsystems with restricted privileges

7. Kernel Mode

- Has full access to hardware and system resources

☒ Key Features of Windows OS

Feature	Description
GUI	Easy navigation using Start Menu, Taskbar, Desktop
Multitasking	Can run multiple applications at once
Virtual Memory	Uses disk as RAM overflow
File System	Uses NTFS for security and performance
Device Management	Plug and Play support
Security	User Account Control (UAC), Windows Defender
Networking	Built-in TCP/IP stack, domain support
Accessibility	Narrator, Magnifier, Speech Recognition
Backward Compatibility	Can run legacy apps
Updates	Delivered via Windows Update

☒ Editions of Windows

Edition	Usage
Home	Personal use
Pro	Professional users, includes domain join, Group Policy
Enterprise	Organizations, enhanced security
Education	Educational institutions
Server	Backend systems and services (e.g., Windows Server 2022)

☒ File Systems Supported

- - FAT32 – Older, compatible with more devices
- - NTFS – Default, supports permissions, compression, encryption
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exFAT – For USB/external drives (large file support)

☒ Security Features

- Windows Defender Antivirus
- BitLocker Drive Encryption
- User Account Control (UAC)
- Windows Firewall
- Secure Boot & TPM support
- Windows Hello (Biometric login)

☒ Task Management & Monitoring Tools

- Task Manager – View and manage processes
- Event Viewer – View system logs
- Performance Monitor – Track system performance
- Resource Monitor – Advanced system monitoring

☒ System Utilities

- Control Panel / Settings

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File Explorer

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Disk Management

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Command Prompt / PowerShell / Windows Terminal

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Registry Editor